Acta Materialia 237 (2022) 118189

Contents lists available at ScienceDirect

Acta Materialia

journal homepage: www.elsevier.com/locate/actamat

Fatigue life estimation of additively manufactured Ti-6Al-4V: Sensitivity, scatter and defect description in Damage-tolerant models

Nicolas Macallister[a,*], Thorsten Hermann Becker[b]

[a] Department of Mechanical and Mechatronic Engineering, Stellenbosch University, Joubert street, Stellenbosch 7602, South Africa
[b] Centre for Materials Engineering, University of Cape Town, Library Rd, Cape Town 7701, South Africa

ARTICLE INFO

Article history:
Received 13 May 2022
Revised 18 July 2022
Accepted 18 July 2022
Available online 19 July 2022

Keywords:
Additive manufacturing
Fatigue strength
Fracture mechanics
Fatigue crack propagation
Multiple defects
Ti-6Al-4V

ABSTRACT

Damage-tolerant models for fatigue life estimation rely on fracture mechanics principles to model fatigue crack initiation and growth. In Laser-Based Powder Bed Fusion (LPBF) produced Ti-6Al-4V defect populations present non-uniform in size, shape, and location within components. Considering defect populations in a statistical nature offers one method to characterise defects to be included in fatigue life modelling for AM parts. In this study a novel numerical fatigue life model is built upon which the sensitivity to input parameters, short crack growth mechanics, defect morphology and location for LPBF-produced Ti-6Al-4V is established. Furthermore, we introduce the potential for multiple crack initiation sites and crack interaction. Fatigue strength is shown to be more sensitive to selected threshold parameters and short crack growth mechanics. Furthermore, the introduction of multiple crack initiations and interaction results in distributions of fatigue strength estimates sensitive to defect number, illustrating the importance of its inclusion as a source of fatigue scatter in LPBF produced Ti-6Al-4V parts. Overall, this study establishes areas of nuance and criticality required in the design and modelling of fatigue life unique to AM produced components.

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1. Introduction

Trusted prediction of fatigue strength in Additive Manufacturing (AM) remains an important hurdle yet to be overcome. Popularised by aerospace and biomedical component production, AM remains one of the promising production paths for building components in near net shape while offering the benefit of low lead times and reduced material waste [1]. One of the more mature process paths under AM is Laser-based Powder Bed Fusion (LPBF) which is widening in suitable application [2]. Of the remaining challenges LPBF faces in fatigue application is either reducing or designing with the melody of process inherent defects formed by the line-by-line and layer-by-layer wise process and rapid melting and solidification fabrication path. The respect in which these defects interact with fatigue strength creates inconsistency in physical test data and mistrust, stemming widespread acceptance into industry [3].

Titanium and specifically Ti-6Al-4V components are common candidates for LPBF owing to their lightweight, high strength mechanical properties [2,4], with the AM material displaying quasi-static properties equivalent to or higher than those produced with traditional manufacturing paths such as casting or rolling. However, this is offset by the lower and inconsistent dynamic/fatigue performance characteristics, which arise from LPBF process inherent artefacts in residual stresses, porosity, surface roughness, and microstructure; further exasperated by the anisotropy in properties between build orientations and machine-to-machine variability [4,5].

Fatigue characterisation of LPBF-produced Ti-6Al-4V remains a prevalent topic as many of the LPBF-produced components are intended for cyclic loading applications [6]. The complex relation between print parameters (laser power, scan speed, hatch spacing and layer thickness), build orientation, surface roughness, inherent defects and residual stresses, and fatigue culminates in significant complexity when establishing fatigue strengths and certification of LPBF-produced Ti-6Al-4V parts for highly critical applications [7]. Moreover, the desire for non-destructive methods to avoid the cost in material and time associated with standardised fatigue testing is becoming increasingly desirable for continued integration of the technique into industry. As such, the state of the art for AM has reached for a predictive approach attempting to train and implement predictive algorithms based on the extensive database of collected physical fatigue data, intending to predict the fatigue strength of LPBF-produced Ti-6Al-4V compo-

* Corresponding author.
E-mail address: 19009143@sun.ac.za (N. Macallister).

https://doi.org/10.1016/j.actamat.2022.118189
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Nomenclature

Abbreviation

AM	Additive manufacturing/manufactured
AF	As -fabricated
AN	Annealed
DA	Duplex anneal
ELI	Extra low interstitial
FCG	Fatigue crack growth
HIP	Hot isostatic pressing
LEFM	Linear elastic fracture mechanics
LOF	Lack-of-fusion
LPBF	Laser powder bed fusion
PBG	Prior beta grain
SR	Stress relief
XCT	X-ray computed tomography

Symbol

Δa	Crack extension
ΔK	Stress intensity factor range
ΔK_{th}	fatigue crack growth rate threshold
$\Delta K_{th,eff}$	Effective fatigue crack growth rate threshold
$\Delta K_{th,lc}$	Long crack growth rate threshold (R dependant)
α	Material three-dimensional stress constraint
σ_{net}	Net section stress
σ_u	Ultimate tensile strength
σ_0	Yield strength
γ	shape parameter
τ	scale parameter
$a_{0,h}$	El Haddad fictitious length scale
C_1	Steady state Paris regime constant
da/dN	Fatigue crack growth rate
d_{10}	10th percentile of powder particle size distribution
d_{50}	50th percentile of powder particle size distribution
d_{90}	90th percentile of powder particle size distribution
F	Velocity factor
l_i	Fictitious length scales
m	Steady state Paris regime constant
R	Load ratio
m	Steady state Paris regime constant
F	Velocity factor
K_{IC}	Mode I plain strain fracture toughness
R_v	Surface roughness valley depth
r_p	Cyclic plastic zone size
U_{LC}	Full formed crack closure parameter
v_i	Weighting factors

nents without destructive testing and extensive standard fatigue specimens.

In description, material defects such as porosity readily present in LPBF-produced components due to the inherent rapid heating and cooling as the part solidifies, which as a bi-product introduces undesirable balling, cracks and porosity type defects [8]. The balling effect has been ascribed to low or high laser energy input, high laser speed or insufficient material, causing a wetting problem of the liquid metal to solid surface, forming irregularly shaped voids in the material [9–11]. Cracks initiate in the solidified material due to the extreme temperature gradients and resultant residual thermal stress produced [12–14]. Porosity type defects form as a result of gas entrapment resulting from hollow material structures, low packing densities and entrapment of evaporated material creating key-hole type pores [10,14]; or as incomplete fusion holes where laser energy input is insufficient for bonding together layers and leaves un-melted powder material between layers [14]. Post-productive steps such as hot isostatic pressing (HIP) have been shown to decrease the size of the internal porosities [15] however it may not entirely eliminate internal porosities and has a lesser effect on defects linked to the surface of the material which heavily influences crack initiation and fatigue behaviour [16].

A popular non-destructive approach for fatigue prediction is adoption of a Damage-tolerant ethos and modelling the fatigue characteristics from a fracture mechanics description of crack growth. This approach which rather assumes the presence of defects as a certainty compared to something to be mitigated entirely, can aid in circumnavigating the problem of inherent defects and allow for utilisation LPBF-produced parts effectively and safely without further thermo-mechanical processing [2]. In application this is often adopted for welded structures wherein accepting the removal of all defects is near impossible, and 'fitness for purpose' is rather a more promising path for design [17]. While many fracture mechanics approaches have been attempted, Madia et al. [18], in reference to welded components, expresses their short fallings in oversimplifying the model, two of which are the use of purely linear elastic fracture mechanics (LEFM) to describe the crack driving force; and secondly the exclusion of the of multiple crack initiation [18]. To effectively portray fatigue life in terms of fracture mechanics instead of traditional approaches such as stress life (S-N) requires altering the fracture mechanics description to consider rather the total lifetime instead of the residual [19]. Zerbst et al. [19] highlight three challenges in this regard to overcome for more accurate use of fracture mechanics in the description of fatigue lifetimes. Namely, these are non-destructive testing (NDT) methods capable of capturing representative initial crack sizes, describing the short crack propagation regime sufficiently and finally modelling crack initiation and growth at multiple locations [19].

Of the non-destructive methods capable of capturing defect position and morphology within a part, X-ray computed tomography (XCT) has become prevalent in the AM environment. The radiographic technique reconstructs defects from multiple penetrating radiation projections [20], allowing for sufficient detection and detail of the most common defects presenting in AM. This becomes further useful in statistical population description of defects to capture quantitative density distributions, which can be incorporated into the AM models. Notably, XCT is limited to size constraints that can fit onto the scan platform and the resolution to detect smaller defect positions and morphology [20]. Further, both contact and non-contact profilometry measurements are commonplace in characterising AM surface asperities, and in modelling approaches linked to the definitions of initial crack size for fatigue [21].

The description of the initiation and propagation of physically/mechanically short cracks within a fracture mechanics framework is not necessarily well captured by linear elastic fracture mechanics. Assuming defects represent initiation sites or idealised cracks and present on a length scale which falls in the transition region between short and long crack growth for LPBF-produced Ti-6Al-4V. In which case, a Damage-tolerant model should emphatically account for the transition into the long crack growth regime where fracture mechanics is traditionally suited. Zerbst et al. [19] identify two mechanisms in this transition region to be considered. The first is the potentially larger plastic zone size ahead of the crack tip relative to the short crack length, which would require description in elastic-plastic fracture mechanics [22]. The second is considering the gradual build-up of the crack closure phenomenon.

The plastic zone size changes as the scale of the crack changes during crack propagation. As the zone ahead of the crack tip represents a region of localised yielding, the simple definition of the crack driving force contained within ΔK may require nuance in altering the basic linear elastic fracture mechanics formulation

to incorporate elastic-plastic fracture mechanics [22]. The most straightforward manner to accomplish this is to add a plastic term to the generic crack length (a) representative of the plastic zone size ahead of the crack ($r_p = 1/6\pi\ (K/\sigma_o)^2$, in a plain strain condition). Alternatively, to increase accuracy of this behaviour is using a cyclic J-integral (ΔJ) as an approximation for the governing crack driving force [23].

The crack closure phenomenon is classified as a contact shielding effect which retards the fatigue damage and therefore crack propagation rate ahead of the crack tip for a portion of the loading cycle [24,25]. This results from a load transfer between the crack faces as they come into contact during the unloading portion of the cycle. Further, closure is classically grouped into roughness, corrosion debris and plasticity induced closure mechanisms. Roughness and corrosion debris closure mechanism form as a result of morphological features on the fracture surface. Where roughness closure forms through local deformation creating asperities on crack faces; and corrosion debris closure through oxidation of exposed crack faces to the atmosphere creating a debris layer during crack propagation [4]. Conversely, plasticity induced closure is ascribed to deformation in the plastic zone ahead of the crack balanced through the plastic out-of-plane flow of material into the crack wake, forming wedges on the crack faces and preventing full crack closure [25]. Focally, plasticity induced closure is the most widely incorporated into fatigue models owed to generality and good empirical description by Newman [26], being conceived from the strip-yield model [27]. However, it may be insufficient to fully describe the crack closure phenomena in LPBF Ti-6Al-4V produced parts, which roughness induced closure is suggested to play a prominent role [28]. The empirical equations proposed by Newman [26] and implemented in Macallister et al. [4] for LPBF-produced Ti-6Al-4V provide a correction value to be applied to the crack driving force (ΔK) to account for the fully developed closure behaviour. This value represents the fully developed crack closure phenomena congruent to and beyond the long crack growth threshold ($\Delta K_{th,lc}$) [29].

Including the closure phenomena with AM inherent defects requires a choice between two paths when describing the inherent defects in a fracture mechanics framework. Either choosing the defects as perfect initial cracks or assuming rather the defects as sharp notches from which fatigue cracks may grow [29]. While description as an ideal crack is simpler, mechanically treating inherent defects as notches includes their morphology's three-dimensional nature meaning a potentially more realistic approach. Therefore, choosing to represent the defects as notches means crack closure values proposed in models such as Newman [26] require first crack extension before becoming fully developed, a crack faces are not initially in contact. It is important to note fully developed closure coincides with the reported $\Delta K_{th,lc}$ values, however physically/mechanically short cracks may still grow from notches at lower ΔK values as a result of the reduced crack closure effects [19,29]. Formulations proposed by Zerbst et al. [19] and Maierhof et al. [29] describe this crack closure build-up. Both approaches use estimates derived from the Kitigawa-Takahashi (KT) diagram, with the difference demonstrated in using the smooth versus exponential approximations for transition between defect-free fatigue strength and long cracks proposed by El Haddad et al. [30] and Chapetti [31], respectively. These formulations are explained in detail elsewhere in [19,29].

Modelling these features for the unique microstructure and defect populations in AM material, raises the complexity of fatigue life predictions. For high cycle fatigue (HCF) the complexity in using the traditional empirical models typical of stress-life numerical simulations increases for AM materials [32]. Where, as a possible answer, probabilistic fatigue assessment approaches to allow for risk and to accommodate fatigue life uncertainties produced by the process induced defects has been used [32]. Successfully applied to LPBF-produced AlSi12 [33] and AlSi10Mg [34], respectively. Taking these complexities for Damage-tolerant design from model stress prediction into account in AM material, this paper sets out to form the critical aspects to consider from a prediction based on defect schema in modelling the fatigue strength of LPBF-produced Ti-6Al-4V. The outcome is to describe fatigue performance better using fracture mechanics and NASGRO formulations, as previously published by the authors [4], and to deterministically account for the scatter in fatigue strength common to AM components. Insight into the dependence of such models on accurate description of mechanically short crack growth regimes, threshold values, and multiple crack initiation sites for LPBF-produced Ti-6Al-4V is to be painted against a backdrop of traditional stress-life (S-N) results for comparison.

2. Numerical methods

The fracture mechanics-based approach presented in this paper optimises the cyclic stress value ($\Delta\sigma$), denoted as fatigue strength in the paper, at a given load ratio (R) required to achieve a specification target number of cycles to failure. The details to the approach, entailing the fatigue crack growth model through NASGRO formulation, multi crack initiation capability, defect population sampling and fatigue strength prediction, are presented below. Where LPBF process inherent defects are equated to crack sizes denoted as (a) and modelled as per the described approach.

2.1. Fatigue crack growth model

A simplified form of the NASGRO solution is presented by Zhang et al. [35] with parameters C_1, p and m established for LPBF-produced Ti-6Al-4V by Macallister et al. [4], further reformulated as suggested in work by Maierhof et al. [29] as,

$$\frac{da}{dN} = C_1 \cdot F \cdot \Delta K^{m-p} \cdot (\Delta K - \Delta K_{th})^{\ p} \tag{1}$$

where, ΔK is the cyclic stress intensity factor, ΔK_{th} the threshold value for ΔK, and the term F represents the crack velocity factor with allowance for crack closure build up as,

$$F = 1 - \left(1 - {U_{LC}}^m\right) \cdot \left[1 - \sum_{i=1}^{n} v_i \cdot \exp\left(-\frac{\Delta a}{l_i}\right)\right] \tag{2}$$

where, U_{LC} represents the fully developed crack closure parameter estimated from Newmans' formulation for plasticity induced crack closure of long cracks [4,26] and the value Δa represents the crack extension from defects during cyclic loading. The values of l_i and v_i represent the fictitious length scales for crack closure formation and fitting parameters for the cyclic R-curve, respectively [29]. The value of n denotes the three closure mechanisms (plasticity, roughness and oxide induced closure), where for this model only plasticity induced closure is modelled, therefore $n = 1$. Fictitious length scale l_1 solely represents the plasticity induced closure mechanism and is estimated from the difference between El-Haddads' fictitious length scale ($a_{0,h}$) for the effective stress intensity factor range ($\Delta K_{eff,th}$) and long crack growth threshold ($\Delta K_{eff,lc}$) recorded in previous work [4,28]. Where $\Delta K_{eff,th}$ represents the intrinsic material crack growth threshold, considered a material property independent of loading ratio [4,28]. To similar effect, the exponential term is included in the threshold build-up description as proposed by Maierhof et al. [29] as,

$$\Delta K_{th} = \Delta K_{th,eff} + \left(\Delta K_{th,lc} - \Delta K_{th,eff}\right) \cdot \left[1 - \sum_{i=1}^{n} v_i \cdot \exp\left(-\frac{\Delta a}{l_i}\right)\right] \tag{3}$$

The proposed constraint $\sum_{i=1}^{n} v_i = 1$, represents the cyclic *R*-curves and considers the build-up of crack closure. The implication of crack closure build-up is considered through a Kitigawa-Takahashi methodology allowing the effects of fatigue strength to be simply visualised. The KT diagram, developed by Kitagawa [36], is a double logarithmic plot depicting fatigue strength ($\Delta\sigma$) at infinite life against crack size. The power of the tool is in marrying the classical fracture mechanics approaches to the traditional stress-based approaches [37]. Further, the point where the horizontal and sloping lines meet represents an estimate of the smallest possible defect to cause initiation.

The model follows a numerical iteration by adding 100 cycles (ΔN) cycles per iteration (j) until the input stress results in survival to the target cycle life. ΔN is chosen sufficiently small relative to total cyclic lifetime with the aim to capture short fatigue crack growth adequately. Crack growth per iteration is determined by the product of crack growth rate per iteration using Eq. (1) and ΔN as,

$$\Delta a = \frac{da}{dN} \cdot \Delta N \tag{4}$$

where the value of crack length (a) for the next iteration (j + 1) is,

$$a_{j+1} = a_j + \Delta a \tag{5}$$

with crack extension (Δa) initially set to zero, analogous with fully undeveloped crack closure. The values for F, K_{th}, ΔK are updated with the current a_j and the total 'survived' cycles on each iteration.

The failure conditions used within the search are either described as survival well beyond the target cyclic life or premature failure, with the premature failure conditions describing the physical fracture of the material. These premature failure conditions are triggered by the net section stress exceeding the material flow stress or ΔK_{max} reaching the plane strain fracture toughness (K_{IC}), using the inequalities described as,

$$\Delta K_{max} \geq K_{IC} \tag{6}$$

$$\sigma_{net} \geq \frac{\sigma_0 + \sigma_u}{2} \tag{7}$$

where, σ_0 and σ_u denote the yield strength and ultimate tensile strength, respectively.

The value for ΔK is implemented into Eq. (1) through the LEFM formulation approximation for crack driving force given as,

$$\Delta K = Y \cdot \Delta\sigma \cdot \sqrt{\pi \cdot a} \tag{8}$$

where, a is derived from Murakami's parameter ($\sqrt{area}$) [38] measured from the modelled defect populations, where the void type defects are initially considered as notches with crack faces not in contact. The stress intensity magnification factor (Y) is initially defined from the closed form solution of a semi-circular surface crack geometry within cylindrical bodies, and allowed to evolve as the fatigue crack propagates. To this extent the impact of defect geometry and location relative to the free surface within the component is accommodated. With the effect of defect location relative to free surface considered through a ratio created considering the Y values of 0.5 and 0.65 suggested by Murakami for internal or near-surface/surface defects respectively [38]. I.e., the closed-form solution used agrees with the Murakami approximation of the initial Y as $\sim$ 0.65, thereafter increasing as the crack grows for a surface defect. However, to differentiate and model internal defects, Y is dampened by a constant factor of 0.5/0.65 to accommodate for a lower stress magnification threat which internal defects pose on the crack tip.

Once a defect is near to the surface, it is transitioned from an internal to near-surface/surface defect through the rule adopted from Murakami [38] as $a/h > 0.8$, where a represents the radius of the equivalent defect and h the ligament distance to the free surface [38].

Finally, the influence of plasticity ahead of the crack tip is included using a simplistic addition to the crack size (a) per Eq. (8), where r_p represents the size of the plastic zone ahead of the crack [39], with a_p replacing the a term in Eq. (8) as,

$$a_p = a + r_p \tag{9}$$

2.2. Multi-crack initiation and defect interaction

Multiple initiation sites and defect interaction present in LPBF-produced Ti-6Al-4V is commonly reported as a necessary consideration in aid of fatigue performance prediction [19]. These effects are incorporated into the fatigue strength prediction model for LPBF-produced parts by populating the representative geometry with representative defect populations (see Sections 2.3 and 3.2), controlling location and size for a given number of defects. Each defect grows independently per iteration.

A coalescence rule on contact is introduced. Within each iteration, coalescence amongst the defect population is identified and interacting defects combined. Defects in contact are combined using a total equivalent area approach where the square root of the sum of the interacting radii is used to be congruent with $\sqrt{area}$-method used throughout the model. Furthermore, the new defect locations on the cross-section are determined using a weighted average of the $\sqrt{area}$ of the interacting defects, from which the defects can be classified as surface/near-surface or internal and accommodated for as described in Section 2.1.

Secondly, independent initiation times and growth rates for each defect based on their size, a, and location, a constituent of Y, are considered. In explanation, different defects express a varied ΔK value based on their independent size and location on the cross-section as per Eq. (8). Therefore, firstly not all defects will necessarily initiate at the same time, and secondly, each may have a different FCG. We, therefore, use an approximation of the growing net section stress to determine when the subsequent defects initiate. The net section stress increases from the applied stress as the largest defect grows through the cross-section resulting from the reduction in area and forming of bending moments. This allows for, as the model progresses, $da/dn > 0$ for less severe defects and initiation to occur, less severe defects typically smaller or internal defects. During each model iteration, da/dN values for each defect are updated and used to determine Δa and a for the next pass, from Eq. (4) and Eq. (5), respectively.

2.3. Extreme value sampling

The estimation of defects considered within the context of Damage-tolerant design becomes a statistical problem depending on the component size and accompanying defect type and distribution [21]. Sound sampling strategies allow for a drastic reduction in considered defects to include in the model by selecting only the most severe defects representative of the most likely possibilities for fatigue initiation and growth. This is important as it removes the numerical effort of considering the entire defect population within LPBF-produced components, without underestimating the effect to which the defect population has on the fatigue strength estimate.

For internal defects, computed tomography (XCT) is commonly used as a tool to generate model defect distributions from a scan of a representative material sample volume [34]. In this regard, sampled extreme values focusing estimates on the upper tail of the

scanned defect distribution is widely adopted as a good measure of failure and growing defect sizes [34]. The peaks-over-threshold (POT) approach offers one method to isolate the extreme tail of defect distributions by describing a threshold value, u, of Murakami's parameter ($\sqrt{area}$) where above-sampled exceedances are considered for fatigue initiation [40]. Choosing a suitable value for u allows for the tail distribution to typically demonstrate asymptotic properties similar to extreme value distributions, from which a Generalised Pareto (GP) distribution can be fitted using the maximum likelihood (ML) method [21].

Romano et al. [34] demonstrated this well for application in the qualification of AM parts, specifically in accurately representing the larger volumes from the smaller scanned volume of material available. This work adopts the formulation to sample defect sizes from the XCT distribution sizes for internal defects and seed the multiple initiation model. The form of distribution function fitted using an maximum likelihood method for determining unique shape and scale parameters γ and τ [40] given as,

$$\begin{cases} W_{\gamma,u,\sigma}(x) = 1 - \left(1 + \gamma \frac{x-u}{\tau}\right)^{-\frac{1}{\gamma}} & for\ \gamma \neq 0 \\ W_{\gamma,u,\sigma}(x) = 1 - e^{-\frac{x-u}{\tau}} & for\ \gamma \to 0 \end{cases} \tag{10}$$

where, x is the distribution of exceedances above u in terms of $\sqrt{area}$. The threshold value u was chosen from a balance between bias and variance, where u is small enough to allow sufficient data points in the tail distribution whilst remaining large enough to allow for the asymptotic nature of the model to be satisfied [41]. Graphically this may be selected [41] considering the mean exceedance curve for the $\sqrt{area}$ XCTdata, according to,

$$Mean\ excess = \frac{\sum_{i=1}^{N_s}(x_i - u)}{N_s} \tag{11}$$

where, N_s represents the number of exceedances and x_i the value of each exceedance of defect size in the distributions obtained from XCT. The value of u is selected where the curve becomes approximately linear, corresponding to a Generalised Pareto distribution that describes the tail distribution [41].

Conversely, for surface aberrations, Baretta [21] suggest that the more common block maxima sampling methodology are better suited for line topographic measurements. The block maxima approach traditionally divides the sample into a series of sub-sections and samples the maximum feature within each sub-section to describe the tail distribution. In the case of surface roughness, these subsections become evaluation lengths or areas measured. Various authors have described these surface asperities as cracks through various roughness parameters as a metric of size. Commonly ten-point height of asperities (R_z), maximum profile height (R_t) and maximum valley depth (R_v) are used [21]. In this study the R_v parameter is selected. Yadollahi et al. [42] suggested R_v is a more realistic representation of initial crack size than other roughness parameters owed to disregarding the exasperated surface asperity height contributed by partially melted powders attached to the surface, as they do not affect fatigue behaviour.

2.4. Fatigue strength prediction

The fatigue strength selection loop is presented in Fig. 1. For each population of defects (internal, near-surface and surface) in the cross-section, a fatigue strength value ($\Delta\sigma$) is selected through a Nelder-Mead simplex algorithm [43]. The optimisation minimises the error between cycles at failure against the target cycle count, which was selected as 10^7 for high cycle fatigue strength representation. As described above, the model checks for defect coalescence on each iteration and updates the location surface/near-surface/internal accordingly as the defects grow.

3. Experimental methods

3.1. Model parameters

The model parameters selected to represent as-fabricated (AF) and annealed (AN) microstructure are summarised in Table 1. In prior work, the FCG rate parameters C_1, p, m and α were established for LPBF-produced Ti-6Al-4V [4]. Similarly, the fatigue long crack and effective threshold values (ΔK_{th} & $\Delta K_{th,eff}$) were established by Becker et al. [28] with the load ratio associated with ΔK_{th} as $R = 0.1$ and $\Delta K_{th,eff}$ as $R \geq 0.7$, where crack closure effects are commonly assumed to subside [44]. Furthermore, defect-free fatigue strength ($\Delta\sigma_e$), plain strain fracture toughness (K_{IC}), ultimate tensile strength (σ_u) and yield strength (σ_0) are selected from reported literature values [2]. To model the AN microstructure, as reported in [45], parameters were selected from a combination of those available in literature [46,47] and found for a duplex anneal (DA) in prior work as per Macallister et al. [4]. Commonly reported within a similar range [2].

3.2. Defect population measurements

For sampling of internal and near-surface defect size, sub-surface defects were measured using XCT scanning on cross-sectional volumes obtained from fatigue specimens for LPBF-produced Ti-6Al-4V in the as-fabricated (AF) surface state (i.e., no material removal affecting near-surface inclusions), schematically represented in Fig. 2. From which defect areas, perpendicular to the modelled fatigue loading in the xy-plane are used in POT extreme value sampling as described in Section 2.3, after further refinement into the desired $\sqrt{area}$ parameter. A threshold value of 50 µm was chosen, as there is some evidence for the curve becoming linear after the initial sharp decay on the mean excess curve at this point in the range (50–70 µm), as seen in Fig. 3a. Furthermore, the estimated model using a generalised pareto fit for the tail distribution with γ and τ fitted as -0.29 and 14.7, respectively (as represented in Fig. 3b) demonstrated goodness-of-fit in both the probability and quantile plots (Fig. 3c and d) with values for adding strength to the threshold choice. The unmodified distribution of defect centres in Cartesian coordinates (xy) converted into polar coordinates (r,θ) from XCT results was used for sampling sub-surface defect locations. Note, no modifications to sample location distribution as the modelled geometry diameter used was equivalent to the cross-section scanned from the specimens.

The second major defect population attributed to detriment of fatigue performance is surface roughness. Measured R_v values were taken using a MarSurf PS10 Perthometer (Mahr GmbH, Göttingen, Germany), which incorporates a Gaussian filter as defined by ISO 16610-21. Five measurements were taken with the travel length parallel to the modelled loading direction. From which maxima R_v values were taken to create a sample pool representative of the most detrimental surface asperities to be modelled as initiating crack sizes as per Fig. 4, in the fatigue life model. The maximum was measured at 31.7 μm for the AF surface in a vertical build orientation.

3.3. Validation fatigue data

Test specimens were printed in a ZXY orientation according to ISO/ASTM52921, designed in a dog-bone morphology according to ASTM E8/E8M-15a and fabricated on an EOSINT M280 (EOS GmbH, Krailling, Germany) LPBF machine. Specimen design was done according to ASTM E8/E8M with a minimum throat diameter ~ 5 mm. Powdered Ti-6Al-4V (ELI) with measured chemical composition constituting of Ti combined with 6.43% Al, 3.94% V,

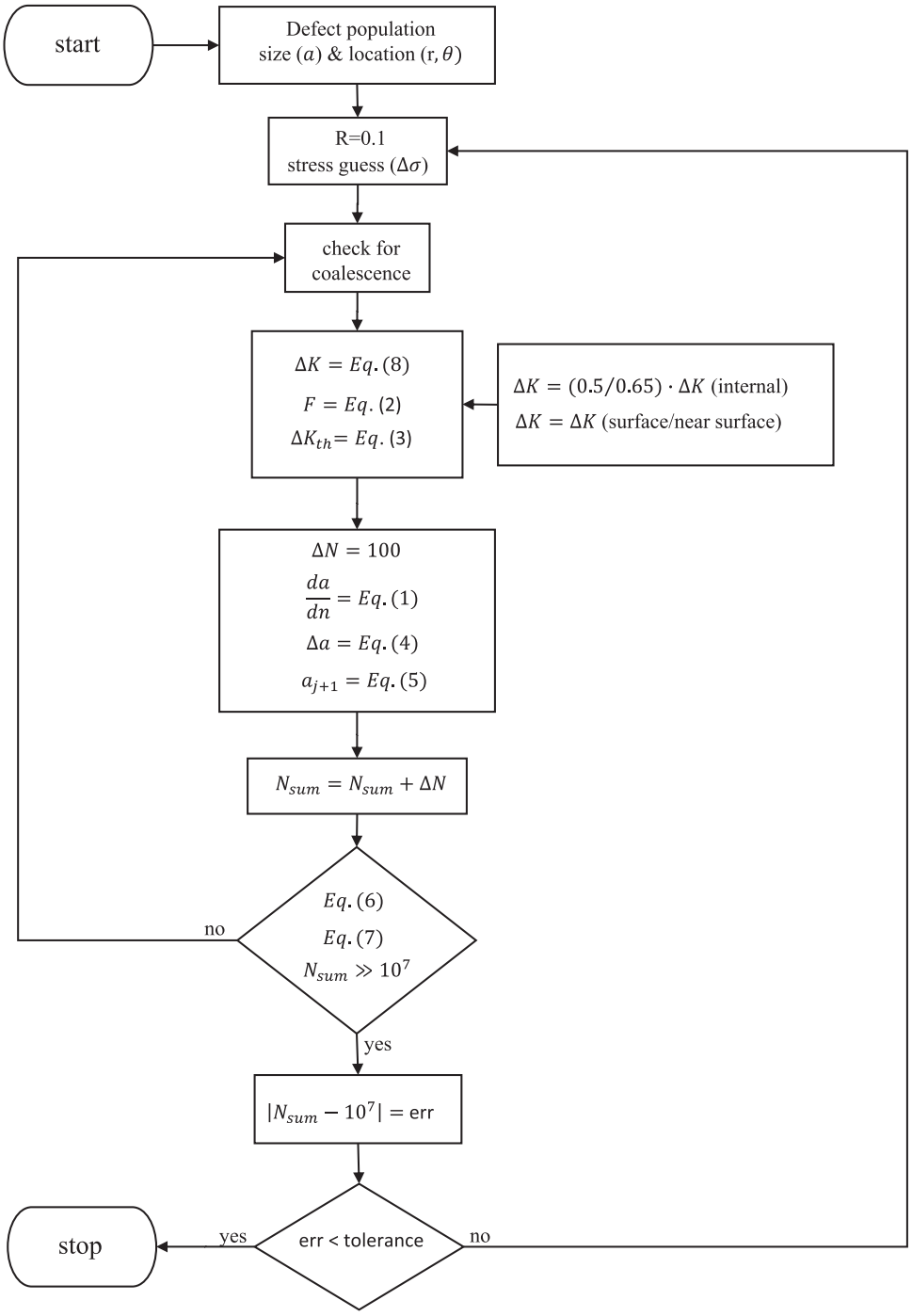


Fig. 1. Model flow for fatigue strength estimate, minimised using the Nelder-Mead method.

Table 1
Model parameters.

	C_1 (m · $cycle/(MPa\sqrt{m})^m$)	p	m	α	ΔK_{th} $MPa\sqrt{m}$	$\Delta K_{th,eff}$ $MPa\sqrt{m}$	$\Delta\sigma_e$ MPa	K_{IC} $MPa\sqrt{m}$	σ_u MPa	σ_0 MPa
AF	1.61×10^{-10}	0.35	2.77	2.73	2.7	1.8	450	50	1200	1100
AN	1.87×10^{-11}	0.35	2.94	1.11	5.6	2.7	487	50	950	900

0.25% Fe, 0.006% N, 0.082% O, with a particle size distribution of $d_{10} = 23$ μm, $d_{50} = 33$ μm and $d_{90} = 46$ μm constituting the raw base material. The parameters for print implemented are presented in Table 2. Argon gas was used to flood the building chamber, and the oxygen level was maintained below 0.12% during the fabrication process. Individual coupons for the AF material state were printed in a near-net shape at a similar height from the baseplate. After separation from the base plate, the density values were confirmed using ASTM B311. Finally, high cycle fatigue (HCF) testing was conducted on a retrofitted Instron 22 kN dynamic testing rig with stress amplitude control at a loading ratio of $R = 0.1$ according to ASTM E466.

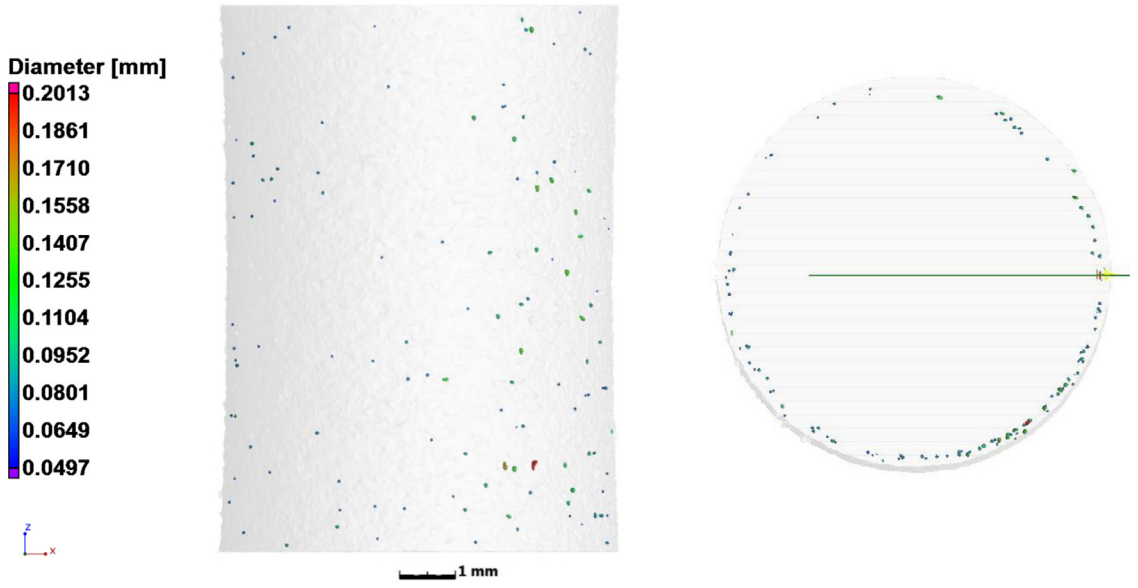


Fig. 2. Subsurface defect population from XCT measurement on fatigue specimen cross sectional volume.

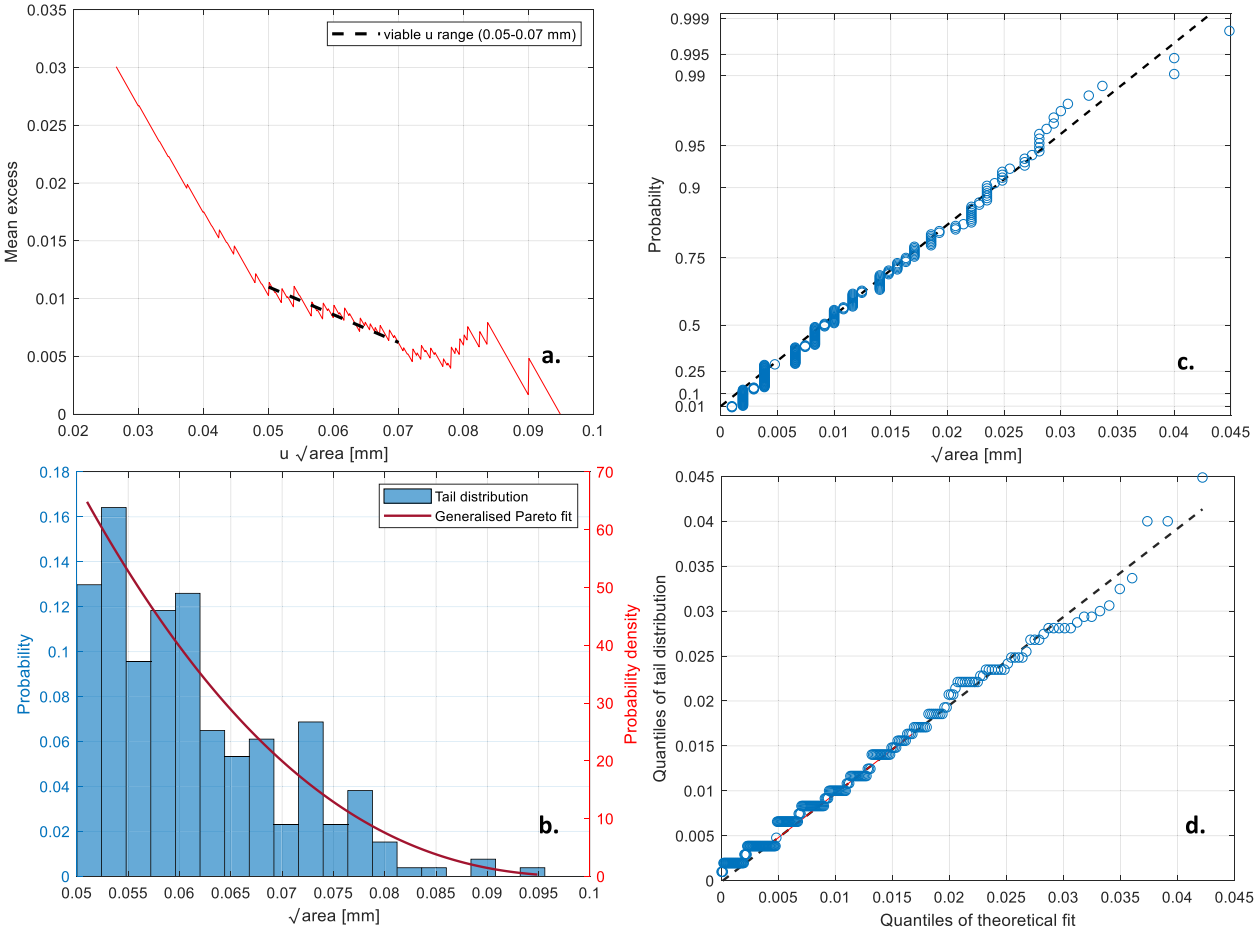


Fig. 3. Extreme value sampling POT (a) mean excess curve with dashed line representing approximate region for suitable threshold selection, (b) normalised probability sampled defect population with fitted extreme value distribution, (c) probability and (d) Quantile-quantile (Q-Q) goodness of fit plots.

Table 2
Printing process parameters.

Power (W)	Scan Speed (mm/s)	Layer Thickness (μm)	Laser Spot Diameter (μm)	Hatch Spacing (μm)	Energy Density (J/mm^3)
170	1200	30	80	100	47

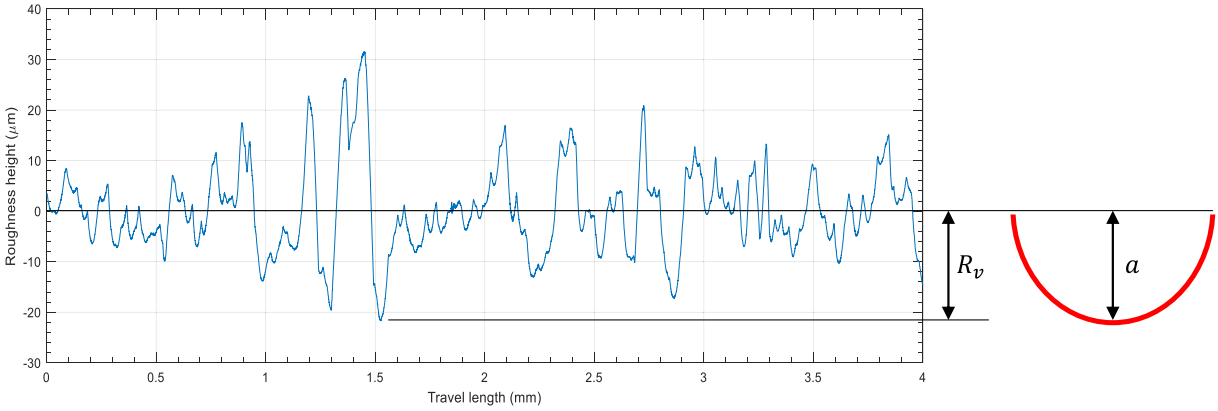


Fig. 4. Surface roughness profile and schematic for equivalent model defect size.

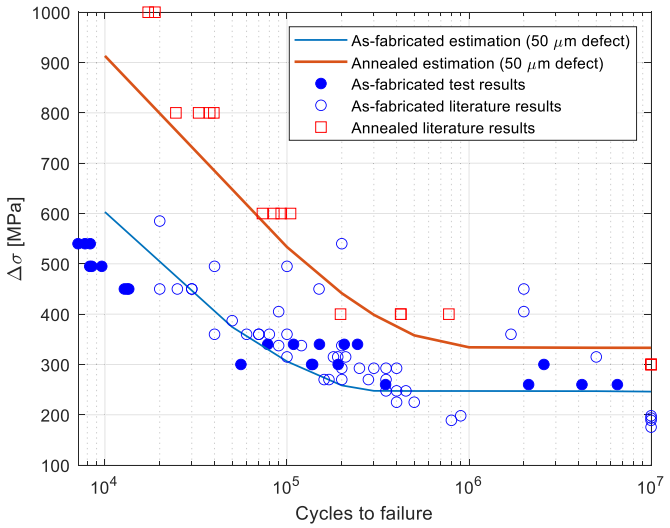


Fig. 5. Model S-N estimates for as-fabricated and annealed unmachined material states.

4. Sensitivity analysis

4.1. Model performance

Firstly, the overall model performance is presented in Fig. 5 in S-N format against the backdrop of fatigue testing data from validation testing and literature for AF [48–52] and AN [45] material both with AF surface finish. Assuming single initiation from surface defect ($a = 50\ \mu$m) at modelled threshold from Section 3.2. The results in Fig. 5 suggest the model performs well within reported scatter in capturing both the S-N shape and scale of reported cycles to failure stress values. Furthermore, by adjusting material and model parameters to represent AN as per Table 1, the model is able to capture better the fatigue performance congruent with an AN heat treatment [45].

4.2. Model parameter sensitivities

Model sensitivity to parameters within the NASGRO formulation, Eq. (1), is presented against the backdrop of literature and the fatigue life data presented in Fig. 6 for AF material. The analysis again uses a single defect with size selected at threshold ($u = 50$ µm) as per Section 3.2 with initiation location at the geometry surface.

Firstly, the impact of variation in extrinsic and intrinsic threshold values, $\Delta K_{th,lc}$ and $\Delta K_{th,eff}$ is presented. These values are significant representations of the early stage of fatigue growth. The results are summarised in Fig. 6a and b with sensitivity to 10% variance around the experimental values of 1.8 and 2.7 for $\Delta K_{th,eff}$ and $\Delta K_{th,lc}$, respectively. The results suggest a more meaningful effect in high-cycle fatigue strength ($> 10^6$ cycles), where decreasing either of the threshold values allows for a higher stress value, with the converse for the increase holding true.

Secondly, the parameter m representative of the steady-state slope (Paris regime) is considered, with the sensitivity of selected stress depicted. The values of m bounded around ±10% from the experimental value of $m = 2.77$. Fig. 6c demonstrates modelled stress values for low-cycle fatigue strength ($< 10^6$ cycles) estimated lower and higher for smaller and larger values of m, respectively. Furthermore, the estimated stress values converge as the cycle count moves towards an infinite life regime (10^7), suggesting the role of the m parameter is more exasperated in low cycle fatigue.

4.3. Plasticity and build-up of crack closure

Allowance for closure build-up and plasticity independently and in tandem within the model is considered to examine their contribution to fatigue prediction. Firstly, their influence on behaviour in an S-N fashion against validation fatigue data is considered in Fig. 7. The addition of a plastic term adjusts the shape of the curve in the low-cycle regime, lowering the stress estimation compared to the model without plasticity. Conversely, the high-cycle asymptote remains unchanged when plasticity is included. Secondly, the addition of build-up of crack closure shows opposing behaviour, instead impacting the high-cycle fatigue strength. Lowering the estimated stress values for the high-cycle asymptote and smoothing and shifting the S-N typical 'knee-point' from $\sim 10^5$ to $3 \cdot 10^5$ cycles.

Further, the effects of the build-up of crack closure is incorporated into the KT methodology (typically used in design for fatigue strength). Fig. 8 shows the KT diagram representation with the build-up of closure both considered and omitted in the model. For KT estimations, the horizontal asymptote representative of the defect-free fatigue strength is chosen from an aggregate of literature values [2], and sloping lines represent the LEFM approximations for a defect-rich material using $\Delta K_{th,eff}$ and $\Delta K_{th,lc}$, respectively. Furthermore, the smooth transition El-Haddad curve is included [30]. Notably, the resulting model curves follow the smooth transition shape of the El-Haddad curve with fatigue strength values, when the build-up of crack closure is included. Superimposed on Fig. 8 is the measured XCT distribution to draw attention to

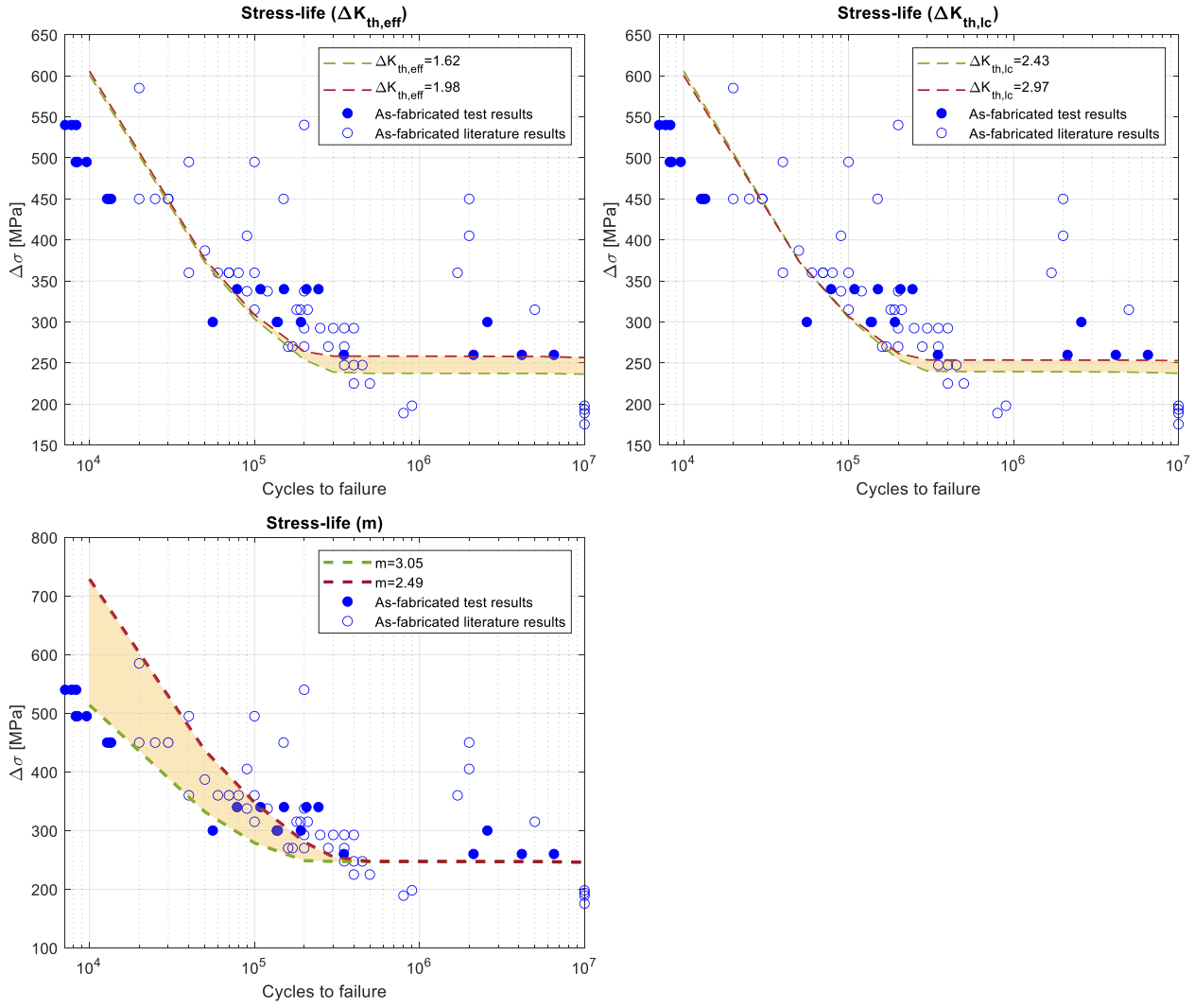


Fig. 6. Model sensitivities to threshold ($\Delta K_{th,eff}$ and $\Delta K_{th,lc}$,) and steady (m) fatigue crack growth parameters.

where typical defects sizes of LPBF-produced Ti-6Al-4V fall relative to the curves.

4.4. Defect location and size

The effect of initiation location is presented in Fig. 9, with upper bound representing a defect positioned at the centre of the modelled geometry. The results demonstrate a decrease in estimated high cycle fatigue strength of 72 MPa (~ 23%) when considering a surface compared to internal (centre) initiation of equivalent defects, with internal defects allowing for higher fatigue strength estimates. Notably, the band between the extreme loci cases is within the reported scatter of fatigue results, suggesting a scatter of crack initiation location relative to the free surface in LPBF-produced components, which may add to the scatter in reported fatigue results. Furthermore, the maximum R_v value sampled from the block maxima technique, as described in Section 2.3, is included in Fig. 9. The relatively lower modelled R_v size of 31.7 μm at the surface when compared to the larger 50 μm internal defect results in a lower fatigue strength estimate.

4.5. Defect population: multiple crack initiation sites

Included in the model is description for multiple crack initiation and growth from defect populations defined in terms of a and the location, r and θ (radial and circumferential, respectively) on the modelled circular cross section. Presented is the effect increasing the number of crack initiation sites has on the modelled fatigue behaviour. To this extent exponentially increasing defect numbers are chosen from eight through to thirty-two and a Monte-Carlo type approach with 100 simulations per defect number is used to determine distributions of fatigue strength estimates. For generating randomly sampled independent defect populations for each simulation: the population of a is selected from the extreme value distribution as described in Sections 2.3 and 3.2, r from a fitted distribution to the measured XCT locations on equivalent cross section, and θ from a uniform distribution on the interval (- π, π). The measured and fitted distributions for the size and radial location are presented in Fig. 10a and b, respectively. To note, surface roughness is not explicitly modelled, however as the radial location sample distribution extends to the surface as shown in Fig. 10b, it is implicitly included as there rests the possibility for surface initiations in each population.

Fig. 11 presents the resultant fatigue strength estimate distributions for each number of crack initiations on a KT backdrop, with the 50th percentile fatigue strength for each denoted by the dashed horizontal line. Where safe life stress denotes the envelope in which the material is not expected to fail. The results demonstrate a shift in fatigue strength estimation to lower values with increasing number of crack initiations, where in comparing the 50th

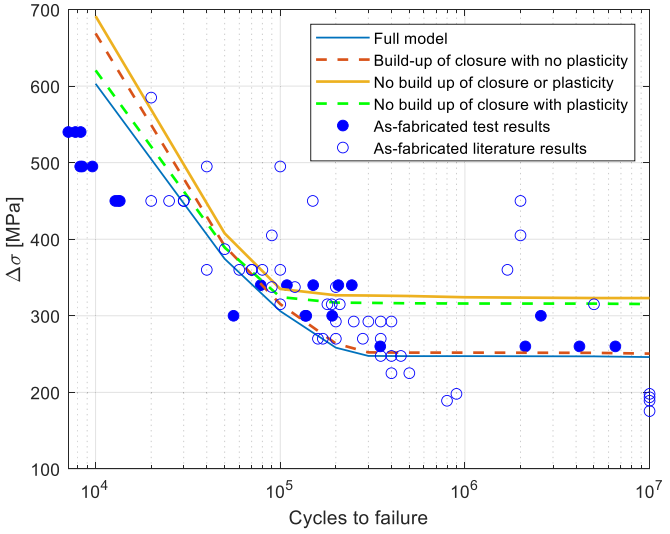


Fig. 7. Effect of physically short crack growth phenomena (build-up of crack closure and plasticity) on S-N model estimates.

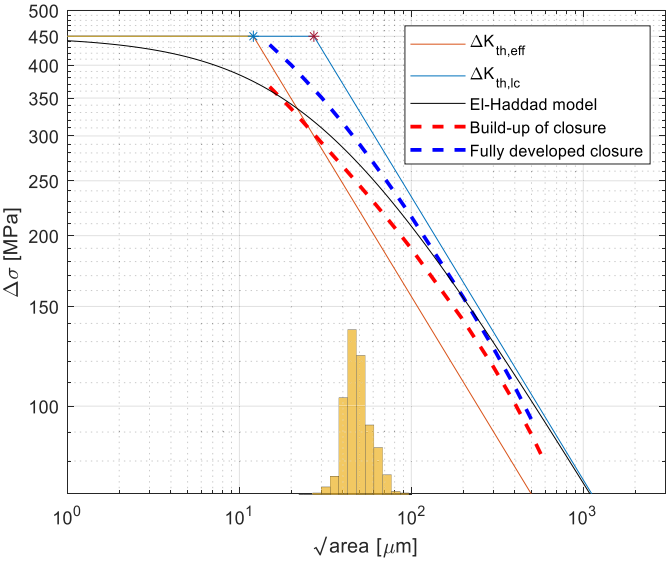


Fig. 8. Effect of physically short crack growth phenomena (build-up of crack closure and plasticity) KT representation.

percentile the fatigue strength is decreased by as much as ∼12% between 8 and 32 initiations.

5. Discussion

5.1. Sensitivity to parameter variation

The sensitivity study in Section 4.1 provides insight into the predicted stresses affected by the variation of parameters within the NASGRO formulation for LPBF-produced Ti-6Al-4V. Paris slope parameter, m, representative of the steady-state propagation regime on a da/dN curve influences low-cycle fatigue prediction without affecting high-cycle prediction. Conversely, the threshold values, $\Delta K_{th,lc}$ & $\Delta K_{th,eff}$, are of greater importance in the high-cycle to infinite life regime, as the initiation stage consumes the majority of cycles at lower stress loadings.

Of concern in qualification and certification of LPBF-produced parts is scatter in reported S-N fatigue and fatigue crack growth data [53]. These may present from the large array of printing parameters, differences between LPBF machines, or testing conditions and methods, to name a few. The result is a range of reported parameters for LPBF-produced Ti-6Al-4V, where prevalent to this research, the values of m are between 2.5 and 4.41 [2], and similarly the load ratio ® dependant $\Delta K_{th,lc}$ has been reported between 1.8 and 6.3 $MPa\sqrt{m}$ [2] for similar build orientation and microstructural state. This suggests for practical implementation of models, such as the NASGRO formulation, to effectively model a 'Damage-tolerant' design approach to better establish the fatigue strength.

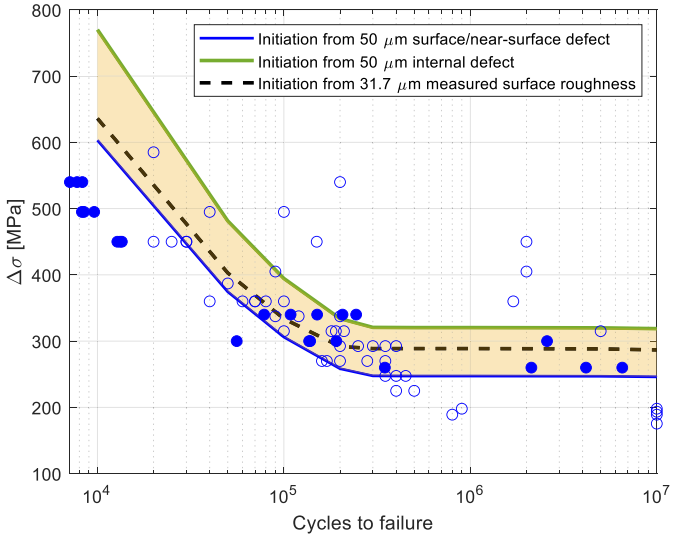


Fig. 9. Size and location effect on fatigue estimates with upper bound representative of defect at centre of geometry.

In application to reliably estimate high-cycle fatigue strength, arguably at a minimum accurate knowledge of $\Delta K_{th,lc}$ & $\Delta K_{th,eff}$ is required to reduce error. This is evident in Fig. 5 whereby increasing these values from 2.7 to 1.8 $MPa\sqrt{m}$ for $\Delta K_{th,lc}$ and $\Delta K_{th,eff}$, respectively, in the AF state, to 5.6 and 2.7 for AN material, results in the high cycle fatigue strength estimation increasing by 26% and capturing the reported improved fatigue performance after heat treatment. Irrespective of the presence of defects ultimately an increase in the ΔK_{th}values allow for a more fatigue resistant material and are critical in high cycle fatigue strength estimation. It also highlights the importance in establishing LPBF process parameters or heat treatments to improve ΔK_{th} for a more Damage-tolerant material.

5.2. Plasticity and build-up of crack closure

Including description for both build-up of crack closure and plasticity are necessary features in describing the propagation of mechanically/physically short cracks in a Damage-tolerant model [19]. The addition of the plastic term into Murakami's $\sqrt{area}$ parameter, as used in this approach, yields notable variability of stress selection in the low-cycle regime, however, negligible impact to the high-cycle regime.

LPBF-produced Ti-6Al-4V in the AF microstructural state is considered a high strength material with a yield strength $(\sigma_0) > 1000$ MPa [46,54–56]. The approximation for r_p ahead of the tip computed as,

$$r_p = \frac{1}{6\pi}\left(\frac{K_{max}}{\sigma_0}\right)^2 \tag{12}$$

in plane strain [57]. The value of r_p is an order of magnitude smaller than the defect sizes during the initiation and short crack growth phase of FCG. Therefore, including plasticity as modelled into the description of crack size suggests a lesser role within

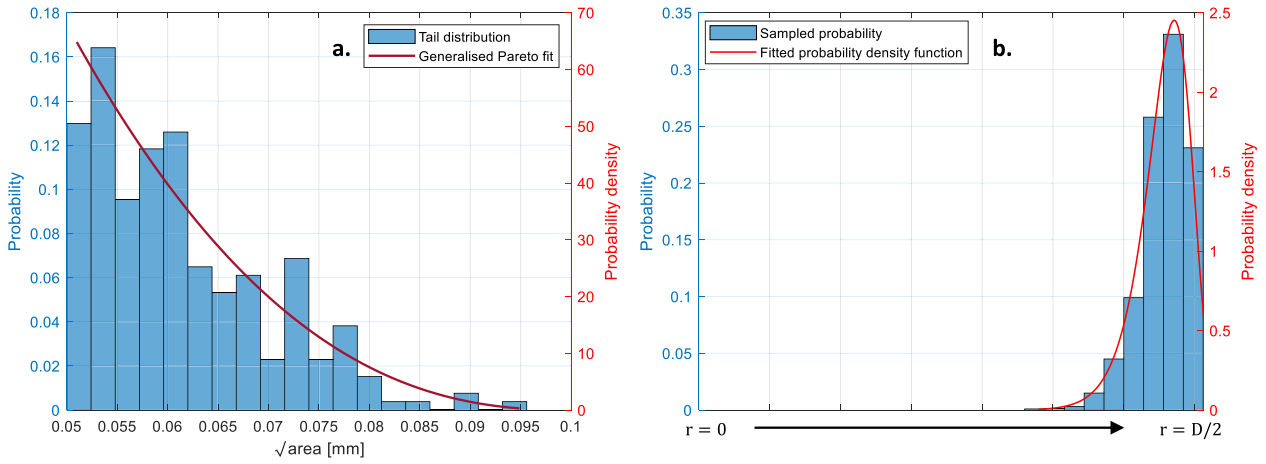


Fig. 10. Multiple initiation model population sampling strategies used for a) size and b) radial location.

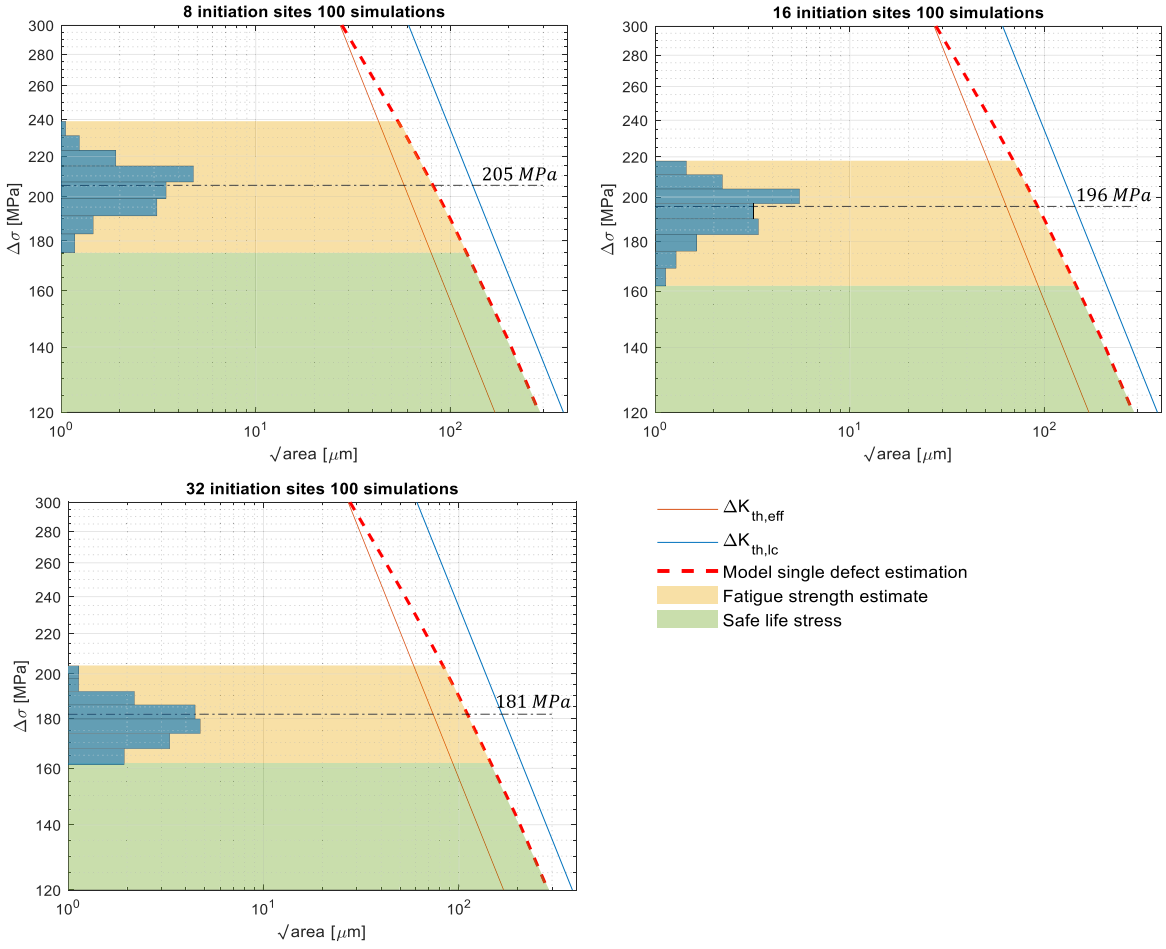


Fig. 11. Multiple crack initiation fatigue strength estimations presented as distributions in KT format, with 50th percentiles represented by the horizontal dashed lines.

the near-threshold region and determination of fatigue strength. However, Kumar et al. [58] suggest the effect of r_p in the near-threshold region contributes rather to the determined ΔK_{th} values as the length scale of r_p at initiation becomes similar to the microstructural features [58]. Where $\Delta K_{th,lc}$ and $\Delta K_{th,eff}$ are shown to affect the estimated fatigue strength in Damage-tolerant models. This suggests that in the case of the AF material, where the plastic zone size is relatively small, the value of including plastic zone size into the description of ΔK for fatigue strength prediction is diminished, and rather its impact is implicit in the threshold values. Notably, this may change for LPBF-produced Ti-6Al-4V with the application of novel heat treatments, aimed at increasing ductility through a microstructural transformation as the values of K_{max} at ΔK_{th} and σ_o both have been shown to change with heat treatment [28,46].

Including build-up of crack closure adjusts the estimated fatigue strength to lower values, as illustrated in Fig. 7. This effect is examined through the relation of crack closure and description

of defects in Damage-tolerant models: Fundamentally crack closure serves as a dampener to FCG by reducing the crack driving force (ΔK) for a portion of the loading cycle as the contact shielding mechanism, constituent of the crack faces in the wake contacting prior to crack tip, absorbs a portion of the load, therefore, retarding crack growth [4]. The value associated with this U_{lc}; if below one, U_{lc} decreases the crack driving force ΔK. U_{lc} is derived from Newman's equations of plasticity induced crack closure [26] for a perfectly flat crack with faces in contact. However, defects in LPBF have faces that are not necessarily in contact due to their porous morphology (opposed to a crack like morphology), meaning that when modelled as a short crack they do not initially experience the same magnitude of dampening congruent with U_{lc}. Therefore, at initiation, crack growth is unhindered by closure effects and therefore accelerated, decreasing when crack faces come into contact, i.e., as the crack extends to a long crack regime. The decreasing FCG rate behaviour is experienced contrary to the crack size increasing within a short crack growth regime. Holistically, this means that the material has a lower fatigue strength. We argue that this is due to the initial and growing size of the defects being sufficiently small not to dominate the estimated FCG rates governed by Eq. (1) in the short crack growth regime. Infinite life is ascribed to cycle consumption in the initiation and near-threshold FCG regimes. Thus, considering the build-up of crack closure becomes important in Damage-tolerant modelling of LPBF-produced Ti-6Al-4V.

Despite progress in refining the LPBF process to reduce porosity, some level of porosity seems to be unavoidable. Thus, the presence of minor defects needs to be considered. Many of the defects fall in the typical range of the transitional region, and we argue that Damage-tolerant models for LPBF-produced Ti-6Al-4V are sensitive to short crack growth mechanics and should be carefully considered.

5.3. Defect morphology and location

The classical LEFM estimation of ΔK Eq. (8) allows for the description of crack size modelled as Murakami's$\sqrt{area}$., however, contains no obvious information regarding the shape of the defect or location relative to the free surface. This information is contained within Y, which evolves as the crack propagates. In this work, the value for Y derives from a simple closed-form solution for surface semi-circular cracks [59], which is a typical approach used in many available Damage-tolerant models. However, this is a simplification as initiating crack shapes in LPBF-produced Ti-6Al-4V almost always exist as irregular shapes and may present below the free surface.

Surface/near-surface defects are more severe on fatigue lifetime than internal defects [42,60]. Leuders et al. [61] suggest that near-surface defects, which were several times smaller than internal agglomerations, dominated the fatigue crack initiations. Lacroix et al. [62] further examined the interaction fatigue crack growth rates and the free surface using an extended finite element method (X-FEM), suggesting FCG rates rapidly increased as subsurface flaws approached the free surface [62]. This can be explained by cracks embedded near a free surface or adjacent cracks experiencing higher stress intensity values in the nearest ligaments due to elevated surrounding stress field [38]. Furthermore, the influence of shape, captured typically through aspect ratio, is suggested to increase the severity to which the subsurface flaws grow through the material ligament and breach the free surface - which may require an additional description to be included in the decision of transforming a subsurface to surface defect [62,63]. To an extent, these are addressed with many fitness-for-service (FFS) codes using simple flaw-to-surface proximity rules and weight-function methodologies to capture the Y value at select defect locations within simple geometries [63].

For LPBF the defect population, measured using extreme value sampling, presents as predominantly near the surface of the specimen (Fig. 2). This suggests the importance of describing the location effect within a model for fatigue strength estimation, as the proximity of defects to both the free surface and one another implies the likelihood of interaction within the cyclic lifetime. Desired application of LPBF-produced components is often in areas of critical loading, which offer complex geometry and stress states. Therefore, models for Y based on simple geometries may introduce errors in fatigue strength estimation. Furthermore, there are various flaw-to-surface proximity rules [63] used within FFS codes which may produce different results. As such, room for part-by-part refinement of stress intensity factor solutions for fatigue life models with complex loadings and geometries may benefit the accuracy of fatigue strength estimation. While the $\sqrt{area}$ parameter may account to an extent for the irregular shape of initiating defects, we argue a degree of nuance is necessary to include in Damage-tolerant models to account for the effects of defect location.

5.4. Surface roughness as a source of fatigue initiation

Describing defect morphologies for fatigue life modelling makes an overarching assumption lumping surface asperities with internal void type defects for numeric simplicity. This, however, is an oversimplification as not all defects are created equally; as is shown in this work, R_v values invariably are smaller than the detected subsurface defects, however, surface asperities are often the dominant initiation sites [64]. Some studies suggest that an as-fabricated surface finish exclusively fails from surface asperities [65].

The measurement and definition of the defect size in modelling for surface asperities as semi-circular geometries with characteristic length a, Fig. 4, may underestimate the effect of surface asperities. Typical 2D surface roughness parameters used in AM such as R_v and R_t [21], typically used as metrics in fatigue life models, likely are limited in accuracy. Common tactile profile measurements, while simpler, are dependent on the size and shape of the probe used in the measurement. An AF surface finish presents with irregular and often partially concealed surface asperities, and tactile measurements invariably may underestimate the severity of the profile.

Non-uniform asperities may extend along the surface (Fig. 12a), meaning the assumption of a simple R_v value describing the crack length may underestimate the severity of the feature. In illustration with our study, if a $\sqrt{area}$ value were to be used measured as per (Fig. 12a) the model value for crack length (a) would transform from our largest measured R_v of 31.7 μm to an equivalent of $\sim$120 μm, which is larger than the 99th percentile of the measured subsurface defect population. This implies the importance of the description of surface asperities; to allow for better competition with near-surface defects in modelling crack initiation.

Additionally, we observe microcracks emanating from the surface (Fig. 12b), which are inaccessible to tactile measuring techniques. These microcracks may exasperate the stress field below the surface and likely influence near-surface defects. As such, surface asperities, near-surface defects, or a combination of both likely govern fatigue initiation and growth and should be modelled uniquely rather than lumped together to be comparable and competitive in fatigue strength estimations.

5.5. Multiple initiations as a source of fatigue scatter

Multiple initiation sites are demonstrated to increase the probability of lower fatigue strength predictions as shown in the re-

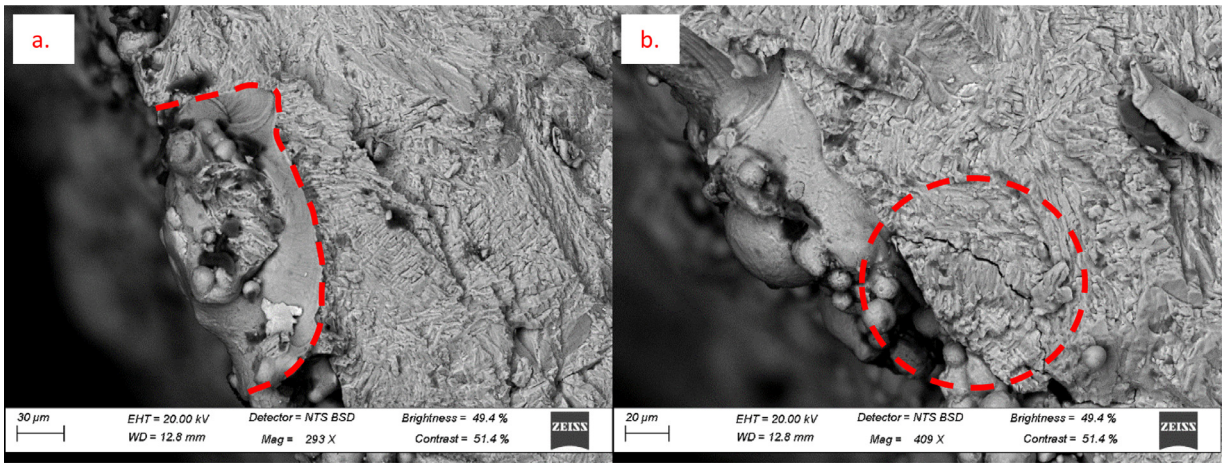


Fig. 12. As-fabricated surface features showing (a) shallow elongated asperities, and (b) micro cracks emanating from surface.

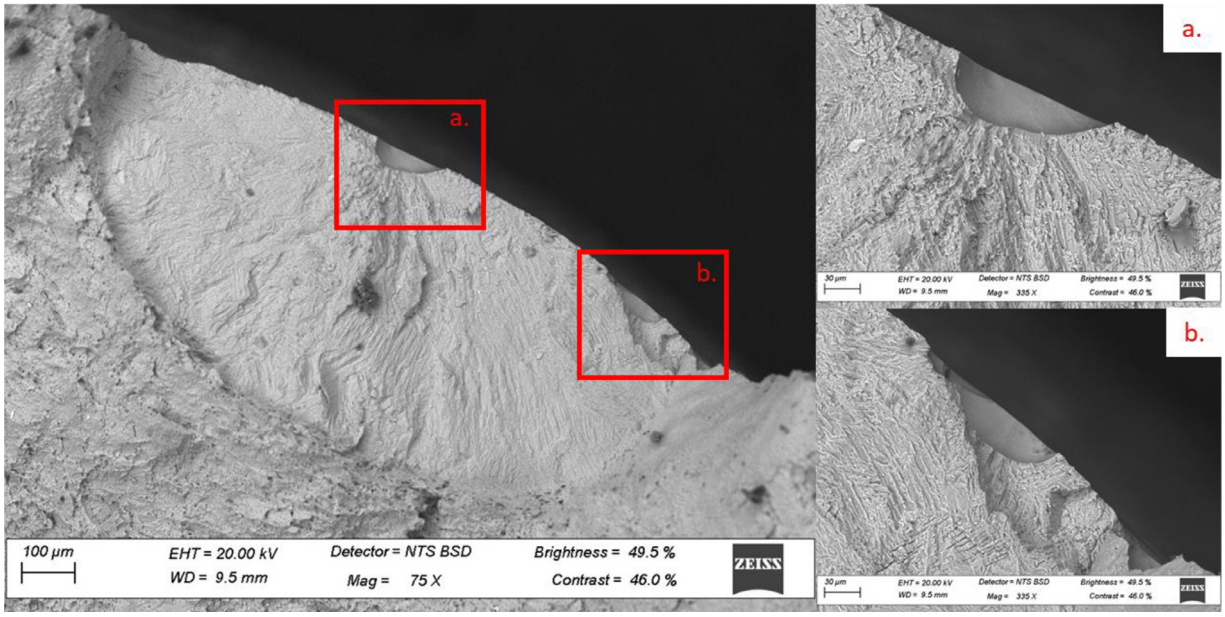


Fig. 13. Multiple initiation sites from exposed defects in a machined surface state.

sults in Section 4.5 (Fig. 11), which makes selecting a design stress difficult for application in LPBF-produced Ti-6Al-4V without being over-conservative. To note, only 100 simulations were performed per Monte Carlo simulation to decrease simulation time, which is typically a small number in the approach, increasing this number may offer better refinement of distributions. Murakami et al. [66] highlights three causes of scatter in fatigue data for AM materials. Firstly, the variability in microstructure is extendable to mesostructure considering the orientation of the prior-β grains between build orientations [58]. Secondly, misalignment of the testing setup and the scatter of defect sizes contained within the material [66]. Furthermore, inherent residual stresses distorting the loading ratio may add to the scatter [28,67]. Within scatter caused by inherent defect sizes, the common assumption considers variance in the morphology and location of a single initiation from the most severe defect in a critical stress area as the culprit for contributing to the scattered lives. The impact and inclusion of multiple crack initiations for LPBF-produced Ti-6Al-4V is less well defined. Fig. 13 presents an illustrative multiple initiation case from a post-mortem analysis on a machined LPBF-produced Ti-6Al-4V fatigue specimen.

Examining weldments, Madia et al. [18] suggest one of the differences between experimental observations and model predictions for weldments boils down to omission of multiple crack propagation in fatigue models, further introducing substantial error in fatigue life prediction [18]. Moreover, the randomness to where the initiation defects are present in weldments translates to vague definition of crack initiation within fatigue life models [68]. LPBF-produced Ti-6Al-4V experiences a similar process path with rapid thermal histories and rough surface finishes, therefore many of the same challenges resulting from the symphony of inherent defects acting as fatigue crack initiations.

Considering the stage of fatigue life where coalescence occurs and the description of the new coalesced crack size is important. Initiation sites near one another will coalesce sooner disrupting the stability of FCG, if this interaction occurs within the near-threshold regime, which is assumed to consume most cycles for near infinite life design, the resultant life prediction will be significantly shorter compared to coalescence within the Paris growth regime. Pegues et al. [69] examined surface roughness within LPBF-produced Ti-6Al-4V, further identifying the relationship between proximity of defects and their impact on stable FCG, with closer initiation sur-

face asperities interacting sooner and leading to less stable growth and shorter fatigue lives. Furthermore, subsurface 'defect clustering' in AM has been suggested to exhibit a similar effect, where smaller clustered defects are potentially more detrimental to fatigue life in comparison to larger defects in isolation [70,71]. Therefore, considering increasing numbers of these initiation defects increases the likelihood that the initial proximity of these growing defects will occur and therefore the probability for lower design stresses allowable, as the distributions in Fig. 11 suggest. Secondly, as the size of the coalesced cracks depends on a weighted estimation of the sizes of the independent interacting cracks, which are non-uniformly ordered through the build geometry, variability between fatigue strength estimations is inevitable, adding further scatter to the fatigue strength estimates. This leads to the necessity of modelling and uniquely considering multiple initiation sites as a source of scatter in damage-tolerant models when designing for fatigue life within LPBF-produced Ti-6Al-4V components.

6. Concluding remarks

The study investigates significant areas of consideration in using Damage-tolerant approaches in application to LPBF-produced Ti-6Al-4V. Evaluating the effect of including nuance in fracture mechanics description of mechanically short crack growth, prediction sensitivity to model parameters representative of the meso- and microstructure and initiating defect morphology and location. Furthermore, the role of surface roughness in AF material and multiple initiation sites as a source of scatter in fatigue results present in AM.

The following conclusions are made:

- The fatigue strength is most sensitive to fatigue threshold parameters $\Delta K_{th,lc}$ and $\Delta K_{th,eff}$ suggesting accurate values are required when Damage-tolerant modelling approaches are used for fatigue estimates.
- The build-up of crack closure impacts stress selection for high-cycle life when defect populations fall within the short crack growth regime.
- Increasing defect size and proximity to the free surface reduces fatigue strength estimates and should be adequately accounted for in Damage-tolerant models.
- Surface roughness in Damage-tolerant models require further refinement and description to represent more accurately severity in fatigue strength estimation accurately.
- Multiple initiation sites reduce the fatigue strength and result in scattered estimates. Therefore, contribute to the scatter of fatigue results reported throughout literature.

Using Damage-tolerant models alongside extreme value statistics to account for scatter in real parts from surface asperities and internal defect populations, provides a useful methodology to better describe fatigue estimates and include space for scatter in LPBF-produced component fatigue strength predictions. Accompanied by a tool such as a KT approach, this may aid in safe fatigue design and more readily certifiable LPBF-produced Ti-6Al-4V builds.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful for the funding from the Department of Science and Innovation through the Collaborative Program for Additive Manufacturing (CPAM) administered by the Council for Scientific and Industrial Research (CSIR), contract number S004063.

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